several kinds of processors in a single system may therefore increase its dynamism and creativity. A conversation between a peasant, a priest and a physician may produce novel ideas that would never emerge from a conversation between three hunter-gatherers.

3. **Increasing the number of connections between processors**. There is little point in increasing the mere number and variety of processors if they are poorly connected to each other. A trade network linking ten cities is likely to result in many more economic, technological and social innovations than ten isolated cities.
4. **Increasing the freedom of movement along existing connections**. Connecting processors is hardly useful if data cannot flow freely. Just building roads between ten cities won't be very useful if they are plagued by robbers, or if some autocratic despot doesn't allow merchants and travellers to move as they wish.

These four methods often contradict one another. The greater the number and variety of processors, the harder it is to freely connect them. The construction of the Sapiens data-processing system accordingly passed through four main stages, each characterised by an emphasis on different methods.

The first stage began with the Cognitive Revolution, which made it possible to connect unlimited numbers of Sapiens into a single data-processing network. This gave Sapiens a crucial advantage over all other human and animal species. While there is a strict limit to the number of Neanderthals, chimpanzees or elephants you can connect to the same net, there is no limit to the number of Sapiens.

Sapiens used their advantage in data processing to overrun the entire world. However, as they spread into different lands and climates they lost touch with one another, and underwent diverse cultural transformations. The result was an immense variety of human cultures, each with its own lifestyle, behaviour patterns and world view. Hence the first phase of history involved an increase in the number and variety of human processors, at the